

JACK GAFFNEY

Herndon, VA · (231) 675-9844 · jackchaudier@gmail.com
jack-chaudier.github.io · github.com/jack-chaudier · Open to SF / NY relocation

SUMMARY

University of Michigan Computer Science graduate and forward-deployed engineer who turns ambiguous, real-world problems into working AI systems. Hands-on across the stack (Python, Go, TypeScript, C++/Rust), with production experience in real-time agent systems, evaluation loops, and multimodal LLM workflows, plus low-level systems work. Best work happens embedded with a team: fast and scrappy under ambiguity, without trading away quality.

EDUCATION

University of Michigan, Ann Arbor, MI

May 2026

Bachelor of Science in Engineering, Computer Science

- **Relevant coursework:** Artificial Intelligence, Operating Systems, Software Engineering, Embedded Systems, Computer Organization, Data Structures & Algorithms, Linear Algebra.

EXPERIENCE

Quantum Opus LLC, Plymouth, MI

Summer 2025

Full-Stack Software Engineering Intern

- Sole engineer embedded with non-technical engineering and operations staff; turned ambiguous, undocumented workflows into validated, shipped decision-support software.
- Designed and deployed a production data platform (Go/Gin, Next.js, PostgreSQL, Docker) that replaced fragmented spreadsheet workflows, iterating directly with users to fit the software to how work was actually done.
- Built ingestion, validation, and entity-linking pipelines to raise data quality and cut manual cleanup across messy internal datasets.

Renewit Decking, Charlevoix, MI

Summer 2024

Carpenter & Logistics Lead

- Scoped projects on site from minimal specification: determined materials, sequencing, and execution with limited direction, adapting quickly across different crews and working styles.
- Promoted from carpenter to logistics lead after identifying recurring bottlenecks; coordinated material scheduling for seven crews across a 50-mile radius to keep simultaneous job sites moving.

SELECTED PROJECTS

PGAI: AI Patient Simulator and Agent Evaluation Loop

2026

Python · Amazon Nova Sonic · Twilio · Gemini

- Built a simulated environment that role-plays patients over live outbound phone calls (Twilio, Nova Sonic) to stress-test a real-time healthcare voice agent under realistic conditions.
- Added a second model (Gemini) as an automated judge to flag failures and pinpoint where the system under test broke down, closing a working evaluation loop for a live agent.

Tether: Real-Time AI Voice System

2026

Go · React · PostgreSQL · AWS Bedrock · WebSockets

- Architected an end-to-end real-time voice system: a Go WebSocket session layer streaming browser audio into Bedrock (Nova Sonic), with PostgreSQL transcript and session persistence.
- Built a post-call LLM pipeline that converts conversations into structured insights and follow-ups, designed to be inspectable rather than black-box.
- Hardened the system for latency-sensitive reliability: session state, transcript fidelity, and failure handling so it held up well beyond a demo.

Vulcan AI: Multimodal Report Pipeline

2026

Python · FastAPI · Deepgram · React

- Owned the voice-agent and report-generation pipeline of a multimodal workflow turning jobsite photos and speech into a formatted PDF report in under 60 seconds (vision, speech-to-text, and LLM synthesis). Hackathon build.

High-Performance Game Engine

2025

C++17 · Lua · SDL2

- Built an ECS-style C++17/Lua engine sustaining 60 FPS at 10,000+ entities under a strict 16 ms frame budget, with custom allocators to prevent heap fragmentation under sustained load.

TECHNICAL SKILLS

Languages: Python, TypeScript, Go, C++, Rust, SQL, PySpark, Lua

AI & Agents: LLM pipelines, model evaluation and failure analysis, fine-tuning, multi-agent orchestration, MCP, real-time voice, multimodal (vision and speech)

Backend & Infra: FastAPI, Go/Gin, PostgreSQL, WebSockets, REST APIs, Docker, Linux, Git, CI/CD, AWS Bedrock

Data: Streaming and large-scale processing, ETL pipelines, validation, structured and unstructured ingestion